

MPC-PiNNs technique applied to Quadrotors

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January 9, 2025

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1 Introduction

outline of the work

unicycle approach

uav approach

2 Model Predictive Control

uav simplified model

uav complete model

mpc structure

3 Physical Informed Neural Network

In our implementation, we have used a Physical Informed Neural Network to predict the next state of the system, given the control input obtained from the Model Predictive Control.

losses

pipeline

image pinn

4 Results

what we have obtained

problems

pinn vs pinnc

5 Conclusions